

Life in harsh environments: carabid and spider trait types and functional diversity on a debris-covered glacier and along its foreland

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Abstract. 1. Patterns of species richness and species assemblage composition of ground-dwelling arthropods in primary successions along glacier forelands are traditionally described using a taxonomic approach. On the other hand, the functional trait approach could ensure a better characterisation of their colonisation strategies in these types of habitat.

2. The functional trait approach was applied to investigate patterns of functional diversity and life-history traits of ground beetles and spiders on an alpine debris-covered glacier and along its forefield in order to describe their colonisation strategies.

3. Ground beetles and spiders were sampled at different successional stages, representing five stages of deglaciation.

4. The results show that the studied glacier hosts ground beetle and spider assemblages that are mainly characterised by the following traits: walking colonisers, ground hunters and small-sized species. These traits are typical of species living in cold, wet, and gravelly habitats. The diversity of functional traits in spiders increased along the succession, and in both carabids and spiders, life-history traits follow the 'addition and persistence model'. Accordingly, there is no turnover but there is an addition of new traits and a variation in their proportion within each species assemblage along the succession. The distribution of ground beetles and spiders along the glacier foreland and on the glacier seems to be driven by dispersal ability and foraging strategy.

5. The proposed functional approach improves knowledge of the adaptive strategies of ground-dwelling arthropods colonising glacier surfaces and recently deglaciated terrains, which represent landforms quickly changing due to global warming.

Key words. Araneae, Carabidae, colonisation, dispersal power, hunting strategies, turnover.

Introduction

Two of the main visible effects of climate warming on alpine areas are glacier retreat and increasing supraglacial debris on glacier surfaces (e.g. Citterio *et al.*, 2007; Paul *et al.*, 2007).

Several studies describe the structural changes (species richness trends and species turnover/persistence) in ground-dwelling arthropod assemblages along the primary succession on recently deglaciated areas (i.e. glacier forelands) (see Hagvar, 2012). Spatial distribution of ground-dwelling arthropods is mainly determined by site age (time since deglaciation), with its related local fine-scale environment conditions, such as soil grain size, vegetation cover, and/or soil organic matter (see, Kaufmann, 2001; Brambilla & Gobbi, 2014; Tampucci *et al.*, 2015). More recently, attention has shifted from the glacier forelands to the surface of debris-covered glaciers, because of the emerging interest of debris-covered glaciers as suitable habitats for micro-, meso- and macro-fauna and plant life (Caccianiga *et al.*, 2011; Gobbi *et al.*, 2011; Azzoni *et al.*, 2015). Debris-covered glaciers are formed by frequent slipping and casting of deposits creating large quantities of stony material which covers the glacier surface, in particular on the ablation area (Citterio *et al.*, 2007), and these have increased significantly during the last decade. Arthropod distribution on debris-covered glaciers is mainly determined by rock grain size, debris thickness, glacier movements/instability, and microclimate conditions (Gobbi *et al.*, 2011).

Traditionally, a taxonomic approach was used to describe ground-dwelling arthropod assemblages along primary successions (e.g. Kaufmann, 2001; Gobbi *et al.*, 2006; Vater, 2012); the functional trait approach has rarely been applied even if it can be useful to understand ecosystem complexity and dynamics (Diaz & Cabido, 2001; Losapio *et al.*, 2015; Moretti *et al.*, 2017). One possible reason might be the lack of knowledge regarding the traits of many taxa and whether these traits are related to environmental changes.

Ground beetles (Coleoptera: Carabidae) and spiders (Arachnida: Araneae) can be considered among the most important meso- and macro-fauna living on recently deglaciated terrains in terms of species richness and abundance (Hagvar, 2012). Carabid beetle and spider life-history traits along environmental gradients (see Schirmel *et al.*, 2012; Pizzolotto *et al.*, 2016) are quite well known in terms of response traits (*sensu* Díaz *et al.*, 2013), i.e. traits that impact on individuals' capacity to colonise and persist in a habitat. On the other hand, there are no studies involving both spider and ground beetle life-history traits that describe, by means of a functional approach, the ground-dwelling arthropod functional diversity and the turnover of life-history traits along a primary succession on glacier forelands and on the glacier surfaces. The functional trait approach would ensure a better characterisation of the arthropod colonisation strategies on the glaciers and on terrain left free by retreating glaciers.

The study area selected to shed light on this topic is one of the few known cases in the Italian Alps in which it is possible to investigate, at the same time, the species assemblage colonisation and the survival strategies on a debris-covered glacier surface and along a more than 160-year glacier foreland

chronosequence. Therefore, first, we describe the carabid and spider species assemblages and life-history traits on the debris-covered glacier, and then we analyse the species richness, life-history trait and functional diversity patterns along the chronosequence of the Holocene glacier retreat. Specifically: (i) we tested whether species richness as well as functional diversity increase with time since deglaciation; and (ii) we hypothesised that time since deglaciation triggers the turnover of life-history traits.

Material and methods

Study area

The study was carried out on the glacier foreland of the Vedretta d'Amola glacier (Adamello-Presanella Group, central-eastern Italian Alps, 46°13'12"–10°41'02") (Fig. 1), and on the glacier surface. Vedretta d'Amola glacier is a debris-covered glacier of c. 82.1 ha (area recorded by one of the authors, RS, in summer 2012), approximately 70% covered with stony debris of variable depth, from a few centimetres to about 1 m. The glacier tongue is located above the treeline at an average altitude of 2730 m above sea level (asl).

The glacier foreland is c. 1.23 km long, covers an altitudinal range of c. 150 m, and is characterised by a large moraine system dating back to the Little Ice Age (LIA, c. AD 1850). Field observations and various sources, including maps, reports, aerial photographs, iconography, and records of length change collected over the last 100 years, allowed us to reconstruct the glacier tongue position during the LIA, in 1925, 1994, and 2003 (Fig. 1).

The snow-free period usually lasts from late June to late September. Annual mean below-ground temperature on the glacier foreland, recorded during the period 5 August 2011 to 5 August 2012, was 1.7 °C, while mean below-ground relative air humidity was 96% (datalogger "A" located at about 15 cm depth in the stony debris at plot 10; see Fig. 1 and Figure S1). The mean annual temperature, recorded during the period 15 July 2011 to 15 July 2012, on the supraglacial debris was 0.5 °C (datalogger "B" located at 10 cm depth in the supraglacial debris at plot 2; see Fig. 1 and Figure S1).

On the supraglacial debris (mean elevation 2642 m asl) the pioneer plant community (total plant cover < 10%) is dominated by *Cerastium uniflorum*, *Cerastium pedunculatum*, and *Saxifraga oppositifolia*. On the glacier foreland (mean elevation 2520 m asl) the plant community (plant cover ranging from 5% to 70% along the foreland) is dominated by *Poa alpina*, *Poa laxa*, *Saxifraga bryoides*, *Geum reptans*, and *Luzula alpino-pilosa*. Outside the glacier foreland (mean elevation 2426 m asl), Late Glacial sites are occupied by *Carex curvula*-dominated communities with > 80% ground cover.

Sampling design

We selected 11 sampling plots located along a linear transect starting on the glacier surface and ending on Late Glacial substrata outside the LIA moraines (Fig. 1). We assigned to each

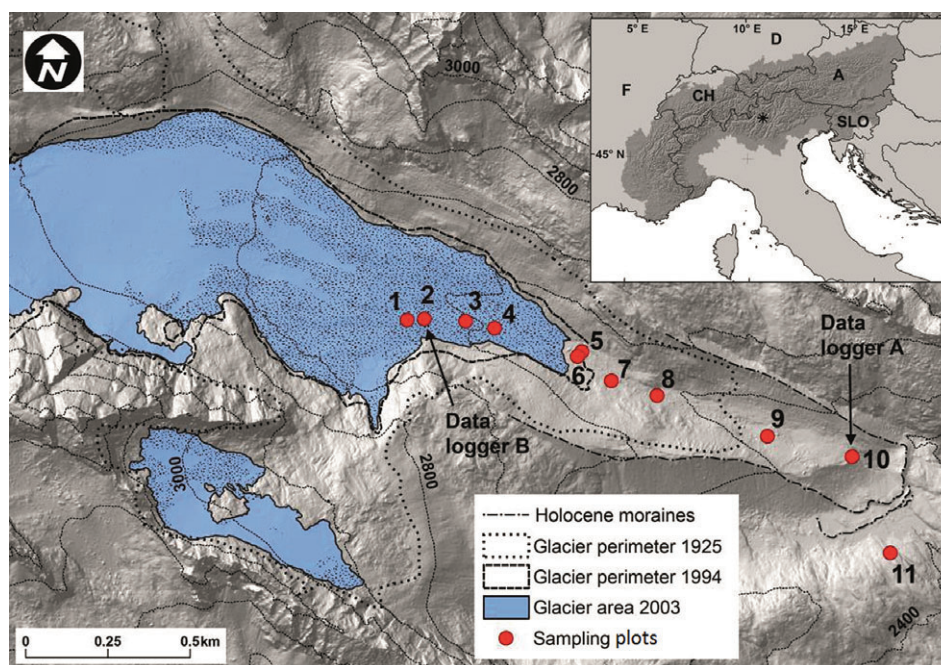


Fig. 1. Geographic location of the sampling plots in relation to the chronosequence of glacier retreat. Plots 1, 2, 3, 4 = class 0 (not yet deglaciated – glacier surface); plots 5, 6, = class 1 (areas freed by the glacier in the period 1994–2003); plots 7, 8 = class 2 (1925–1994); plots 9, 10 = class 3 (1850–1925); and plot 11 = class 4 (Late Glacial Period). [Colour figure can be viewed at wileyonlinelibrary.com].

plot a class of deglaciation: class 0 (not yet deglaciated – glacier surface; plots 1, 2, 3, 4); class 1 (areas deglaciated in the period 1994–2003; plots 5, 6); class 2 (1925–1994; plots 7, 8); class 3 (1850–1925; plots 9, 10); and class 4 (Late Glacial Period; plot 11) (Fig. 1). Plots were selected on the basis of the following two criteria: (i) areas not subjected to physical disturbance (e.g. rockslides, river flooding); and (ii) detection probability of the considered species (e.g. on the glacier surface, class 0, we located four plots due to the low species detection probability; see Tenan *et al.*, 2016).

Sampling method

We sampled carabid beetles and spiders using pitfall traps (Eymann *et al.*, 2010). In each plot three traps were located about 10 m apart (Kotze *et al.*, 2011), which led to a total of 33 pitfall traps. Traps consisted of plastic vessels (diameter 7 cm, height 10 cm) baited with a mixture of wine-vinegar and salt. The traps were active over the entire snow-free seasons, from early July to late September 2011–2012. Samples were taken at 25-day intervals. Plots 2, 4, 5, 8, and 10 were studied in year 2011, whereas plots 1, 3, 6, 7, 9, and 11 were studied in 2012 in order to optimise the sampling effort in this kind of harsh environment. As temperature is one of the main factors affecting carabid life cycle, distribution, and species assemblage composition in montane habitats (Kotze *et al.*, 2011), the choice to sample in two different sampling years could be a bias in our dataset. Thus, we compared the annual mean temperature in each of the sampling years (2011 vs. 2012). We considered air temperature data from the nearest

(c. 6.5 km from our study site) meteorological station (Pinzolo, Italy; latitude $46^{\circ}09'22''$ – $10^{\circ}45'25''$, elevation 760 m asl; www.meteotrentino.it). The mean annual temperature in 2011 was 9.2°C , while it was 9.0°C in 2012. Given that no important variation in air temperature was found, we can assume that it did not affect the carabid beetle assemblage richness and composition in the two sampling seasons.

Carabids were identified to the species level following Pesarini and Monzini (2010, 2011), while spiders were identified to the species level following (Nentwig *et al.*, 2017).

Nomenclature refers to the checklist of the European Carabid Beetles Fauna (Vigna Taglianti, 2013) and to The World Spider Catalogue (World Spider Catalog, 2017). For spiders, juveniles were excluded from the analysis.

Environmental variables

We recorded abiotic (percentage of gravel, soil pH, and soil organic matter) and biotic (plant species richness and vegetation cover) variables within a buffer of 1 m around each trap.

We collected a substrate sample of 1–2 kg at every plot for particle size distribution. In all, 200 g of substrate were sampled at each pitfall trap for organic matter content analysis (Walkley–Black method: Walkley & Black, 1934) and pH measurement. All the soil samples were taken at the surface. We recorded plant cover using a 50-cm-diameter metal circle placed at the four opposite sides of the pitfall trap. We recorded vascular plants, bryophytes, and ground lichens occurring within the plot and visually estimated the overall vegetation cover and that of every species, with a resolution of 5%. We calculated the

mean values from the four 50-cm samplings to obtain a single value associated with each trap. For each plot, we recorded and averaged environmental variables around the three pitfall traps.

Functional traits

Carabids. We considered the following well-established response traits of primary succession (Gobbi *et al.*, 2010; Schirmel *et al.*, 2012): dispersal ability (high dispersal power = winged species; low dispersal power = short-winged species); diet [omnivorous (*sensu* Talarico *et al.*, 2016), carnivorous]; and mean body length (mm) of the pool of species in each trap. We analysed for the first time along primary succession the following traits: larval hunting strategy (surface runner, surface walker, soil pore explorer) and adult hunting strategy (zoospermophagous plant climbers, olfactory-tactil predator, visual predator). Data about species traits were obtained on the basis of specialised literature (Hürka, 1996; Bell *et al.*, 2005; Brandmayr *et al.*, 2005; Blandenier, 2009; Homburg *et al.*, 2014; Talarico *et al.*, 2016; Nentwig *et al.*, 2017) (Table 1).

Spiders. We considered the following response traits: adult dispersal ability (flying dispersers = ballooners; ground dispersers = walkers); hunting strategies (ground hunters, sheet web weavers, other hunters); and mean body length (mm) of the pool of species in each trap. Traits were gathered on the basis of Nentwig *et al.* (2017) and specific information on ballooning was derived, whenever possible, from the literature (Bell *et al.*, 2005; Blandenier, 2009). We assigned functional groups according to Cardoso *et al.* (2011). In this respect, the mixed guild 'other hunters' – small sheet web weavers and stalkers – includes, in our case, Linyphiidae belonging to the subfamily Erigoninae (Salticidae are represented by one single species, and two individuals) (Table 1).

For each trap, we calculated the proportion of each trait within the community.

Data analysis

Environmental variables. Due to the high number of environmental variables recorded in the field, we performed a preliminary correlation analysis in order to minimise multicollinearity-related problems on the estimate of the regression model parameters (Legendre & Legendre, 2012) and to test if they are a function of the time since deglaciation (class of deglaciation). Time since deglaciation, vegetation cover, plant species richness, soil gravel percentage, pH, and organic matter turned out to be highly correlated (Spearman's $\rho > 0.9$ in all cases except one – time since deglaciation versus pH – where it is > 0.7 ; Table S1). Thus, on the basis of all previous information suggesting the importance of time since deglaciation, the latter was entered as the sole explanatory variable in statistical models. This choice was further motivated by the fact that time since deglaciation is the only variable that may influence the others, with a positive influence on plant species richness, percentage of vegetation cover, organic matter

content and a negative influence on pH and gravel percentage. Furthermore, we described time since deglaciation as the main variable influencing carabid and spider primary succession along glacier forelands (see Hagvar, 2012).

Diversity. Species richness was expressed as the number of species per trap (count data). According to Mason *et al.* (2013), we computed the index of functional richness (FRic) and functional dispersion (FDis) as descriptors of the functional diversity of carabid and spider assemblages along the succession. These two functional diversity indexes are indicators of community assembly processes (Mason *et al.*, 2012). Functional richness measures how much of the niche space is occupied by the species present. It is usually interpreted as an indicator for potentially used/unused niche space (Schleuter *et al.*, 2010). Functional dispersion estimates the dispersion of the species in the multidimensional trait space, calculated as the weighted mean distance of individual species in the trait space to the weighted centroid of all species, accounting for species relative abundance (Laliberté & Legendre, 2010). In plots 1 and 2 of the class 0, no carabids were collected; thus, FDis were excluded when calculating FDis for the carabid community in these plots.

Species traits distribution. We analysed the turnover or 'persistence' of carabid and spider traits along the succession according to the descriptive analysis proposed by Vater (2012) and Vater & Matthews (2013, 2015). Specifically, we analysed three community parameters for each class of deglaciation: (i) total functional traits (number of functional traits at plot level); (ii) first appearances of functional traits (number of functional traits appearing for the first time along the succession, including first and last appearances); (iii) last appearances of functional traits (number of functional traits appearing for the last time along the succession, including first and last appearances).

Statistical analysis. Given that our data have a clear spatial structure, with three traps within each sampling plot, and that spatial autocorrelation is a key issue for studies investigating invertebrate ecology (and carabids and beetles in particular) along glacier forelands (Gobbi & Brambilla, 2016), we adopted a modelling technique able to deal with spatially autocorrelated data. We worked with generalised least squares (GLS) models, which can incorporate the spatial structure into the model's error and are one of the best performing methods for similar spatial analyses (Dormann *et al.*, 2007; Beale *et al.*, 2010). We thus used GLS models to estimate the potential effect of time since deglaciation on the selected traits/indexes, and checked for residuals distribution for all models for which the effect of time since deglaciation was not rejected; in all but one (proportion of winged species among ground beetles) of such cases, residual distribution approached a normal distribution. We assessed models' support by means of an information-theoretic approach (Burnham & Anderson, 2002), based on Akaike's information criterion corrected for small sample size (AICc): in all cases when the model including the factor time since deglaciation was better supported than the null model, we

Table 1. Carabid and spider species assemblages and life-history traits in each class of deglaciation (class 0 = not yet deglaciated – glacier surface; class 1 = areas deglaciated in the period 1994–2003; class 2 = areas deglaciated in the period 1925–1994; class 3 = areas deglaciated in the period 1850–1925; and class 4 = Late Glacial Period).

	Class 0	Class 1	Class 2	Class 3	Class 4	Dispersal ability	Adult hunting strategies	Larval hunting strategies	Diet	Mean body length (mm)
<i>Amara erratica</i>			1.2	1.7	1.1	High	Zoospermophagous plant climber	Spermophagous	Omnivorous	7.2
<i>Carabus adamellicola</i>			0.4	10.0	5.6	Low	Olphactory tactil predator	Surface walker	Carnivorous	19
<i>Carabus depressus</i>			0.8	11.7	6.1	Low	Olphactory tactil predator	Surface walker	Carnivorous	22.5
<i>Cychrus attenuatus</i>			0.4			Low	Olphactory tactil predator	Surface walker	Carnivorous	15
<i>Nebria germari</i>	100.0	92.6	39.3	2.2	0.6	Low	Olphactory tactil predator	Surface runner	Carnivorous	10.25
<i>Nebria jockischii</i>		5.6	0.8			High	Olphactory tactil predator	Surface runner	Carnivorous	12.2
<i>Notiophilus biguttatus</i>				0.4	1.7	Low	Visual predator	Surface runner	Carnivorous	5
<i>Oreonebria angustata</i>		1.9	12.3	0.4		Low	Olphactory tactil predator	Surface runner	Carnivorous	8
<i>Oreonebria castanea</i>			43.4	64.9	76.0	Low	Olphactory tactil predator	Surface runner	Carnivorous	8.8
<i>Platynus teriolensis</i>				0.9	8.9	Low	Olphactory tactil predator	Surface walker	Carnivorous	11.25
<i>Principidium bipunctatum</i>			0.4	0.4		High	Olphactory tactil predator	Soil pore explorer	Carnivorous	4
<i>Pterostichus multipunctatus</i>			0.4	7.4		Low	Olphactory tactil predator	Soil pore explorer	Carnivorous	14
<i>Trechus tristriculus</i>			0.4			Low	Olphactory tactil predator	Soil pore explorer	Carnivorous	4
		Class 0	Class 1	Class 2	Class 3	Class 4	Dispersal strategies of the adult	Hunting strategies		Mean body length (mm)
<i>Acantholycosa pedestris</i>				3.8			Ground disperser	Ground hunter		9.25
<i>Agyneta rurestris</i>		77.8				11.1	Ballooners	Sheet web weaver		2.18
<i>Arctosa alpigena</i>						11.1	Ground disperser	Ground hunter		6.80
<i>Coelotes pickardi tirolensis</i>				30.8	52.9	55.6	Ground disperser	Sheet web weaver		8.85
<i>Diplocephalus helleri</i>			25.0	3.8	5.9		Ballooners	Other hunter		2.18
<i>Drassodex heeri</i>					20.6		Ground disperser	Ground hunter		18.15
<i>Erigone dentipalpis</i>			25.0	3.8			Ballooners	Other hunter		2.33
<i>Mughiphantes handschini</i>				15.4	2.9		Ballooners	Sheet web weaver		2.75
<i>Oreonetides glacialis</i>			25.0	3.8			Ballooners	Sheet web weaver		2.68
<i>Pardosa nigra</i>	22.2	25.0	26.9	11.8	22.2		Ground disperser	Ground hunter		7.75
<i>Pardosa oreophila</i>			7.7				Ground disperser	Ground hunter		5.55
<i>Sitticus longipes</i>					5.9		Ground disperser	Other hunter		9.15
<i>Tenuiphantes monachus</i>				3.8			Ballooners	Sheet web weaver		2.60

Specimen abundance of each species is indicated as a percentage of total captures in each class of deglaciation. Data about carabid and spider life-history traits were obtained on the basis of specialised literature (Hürka, 1996; Bell *et al.*, 2005; Brandmayr *et al.*, 2005; Blandenier, 2009; Homburg *et al.*, 2014; Talarico *et al.*, 2016; Nentwig *et al.*, 2017).

considered time since deglaciation as a meaningful predictor of a given trait/index; otherwise, we treated it as uninformative for such a parameter. We run models using three different correlation structures (Gaussian, spherical and exponential; see, e.g., Brambilla & Ficetola, 2012) and obtained fully consistent results between the three runs.

The proportion of adult carabid hunting strategies with each class of deglaciation were not tested by the GLS as all the species except two [*Amara erratica* (Duftschmid 1812) and *Notiophilus*

biguttatus (Fabricius 1779)] are olphactory-tactil predators, in the same way that adult feeding guilds were not tested as all species except one (*Amara erratica*) are carnivorous (see, Table 1). The proportion of carabid species with omnivorous larvae was not tested by the GLS as all species except one are carnivorous, and thus we tested only the proportion of omnivorous individuals in each site. The spider hunting guild ‘other hunter’ proportion was not tested by the GLS because it belonged only to three species and eight individuals.

Table 2. Summary of the effect of time since deglaciation (class) on the response variables (species richness, species traits, and functional indices).

Carabids	Intercept	Class 1	Class 2	Class 3	Class 4
Species richness	0.41 ± 0.43	0.92 ± 0.72	3.92 ± 0.72***	4.60 ± 0.72***	4.92 ± 0.93***
Proportion of surface walker larvae	0.00 ± 0.05	0.00 ± 0.08	0.09 ± 0.07	0.28 ± 0.07**	0.50 ± 0.09***
Proportion of surface runner larvae	1.00 ± 0.09***	0.00 ± 0.15	−0.25 ± 0.13	−0.53 ± 0.13	−0.59 ± 0.17***
Proportion of soil explorer larvae	0.00 ± 0.04	0.00 ± 0.06	0.09 ± 0.05	0.18 ± 0.06**	0.00 ± 0.07
Proportion of winged species	0.00 ± 0.06	0.38 ± 0.09***	0.20 ± 0.08*	0.11 ± 0.08	0.11 ± 0.09
Mean body length	10.25 ± 0.84	0.49 ± 1.34	0.43 ± 1.17	2.07 ± 1.17	2.06 ± 1.55
FRic	1.00 ± 0.48	0.78 ± 0.75	1.48 ± 0.65*	2.31 ± 0.67**	2.03 ± 0.87*
FDis	0.00 ± 0.04	0.04 ± 0.05	0.07 ± 0.05	0.11 ± 0.05*	0.10 ± 0.06
Spiders	Intercept	Class 1	Class 2	Class 3	Class 4
Species richness	0.73 ± 0.31*	−0.06 ± 0.52	1.94 ± 0.52***	1.77 ± 0.53**	2.63 ± 0.70***
Proportion of ground hunters	0.17 ± 0.13	0.08 ± 0.21	0.32 ± 0.19	0.20 ± 0.22	0.33 ± 0.26
Proportion of sheet web weavers	0.83 ± 0.12	−0.58 ± 0.20**	−0.38 ± 0.18*	−0.42 ± 0.20*	−0.33 ± 0.24
Proportion of ballooning	0.83 ± 0.13***	−0.07 ± 0.20	−0.47 ± 0.18*	−0.66 ± 0.20***	−0.63 ± 0.25*
Mean body length	3.11 ± 0.84***	0.61 ± 1.33	2.86 ± 1.20*	5.70 ± 1.35***	3.80 ± 1.62*
FRic	1.33 ± 0.28***	0.66 ± 0.44	1.50 ± 0.39**	1.67 ± 0.41***	3.00 ± 0.48***
FDis	0.16 ± 0.06	0.16 ± 0.10	0.23 ± 0.09*	0.32 ± 0.09*	0.31 ± 0.11*

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

The cases for which the effect of time since deglaciation was supported (model including the variable more parsimonious than the null model; see text) are reported in bold; for all other variables, the model with time since deglaciation was less supported than the null model. Values are estimated coefficients ($\pm$ relative standard error) for the effect of time since deglaciation in relation to class of deglaciation 0 (glacier surface); also, the significance of effect is tested against class 0. FRic, functional richness; FDis, functional dispersion.

All statistical analyses were performed with the software R (R Development Core Team, 2016), using FD R package version 1.0-12 (Laliberté *et al.*, 2014) to compute the functional diversity indices and the packages ‘MuMIn’, ‘mass’ and ‘nlme’ (Venables & Ripley, 2002; Bartoń, 2016; Pinheiro *et al.*, 2017) for GLS models.

Results

Diversity trends

A total of 13 carabid species (732 individuals) and 13 spider species (91 individuals) were sampled (Table 1).

Carabid and spider species richness increased along the chronosequence of glacier retreat as described by the four classes of deglaciation ($R_{\text{carabids}}^2 = 0.72$; $R_{\text{spiders}}^2 = 0.57$). For both carabids and spiders, the species richness value was low on the glacier and during the early successional stage (class 1), and then it increased, but not linearly (Table 2, Fig. 2).

None of the two functional diversity indexes of carabid assemblages used was correlated to the time since deglaciation. Spider FRich gradually increased in relation to the time since deglaciation ($R^2 = 0.71$) and with a similar trend observed for the species richness (Table 2, Fig. 3). On the other hand, spider FDis did not change in relation to the time since deglaciation.

Life-history trait proportion

Among carabids, the proportion of surface walker larvae gradually increased along the chronosequence of glacier retreat as

described by the four classes of age of deglaciation ($R^2 = 0.78$) (Table 2, Fig. 4). Conversely, the proportion of surface runner larvae ($R^2 = 0.73$) gradually decreased in relation to the time since deglaciation. Instead, the proportion of soil explorer larvae did not change along the chronosequence. The proportion of high dispersal species reached the highest values in the early successional stages (class 1), then gradually decreased along the chronosequence of glacier retreat ($R^2 = 0.70$) (Table 2, Fig. 4). The community-weighted mean body length of the species in each trap did not change along the chronosequence.

Among spiders, the proportion of ballooning species reached the highest value on the glacier surface, then it decreased along the chronosequence of glacier retreat as described by the four classes of age of deglaciation ($R^2 = 0.56$) (Table 2, Fig. 3). The proportion of each hunting strategy did not change along the chronosequence. The mean body length of the species pool in each trap increased along the chronosequence of glacier retreat ($R^2 = 0.67$) until the class 3, and then it decreased slightly (Table 2, Fig. 3).

Life-history trait distribution

For carabids, the total number of functional traits increased among classes of deglaciation until the class 3, and then slightly decreased in class 4. Functional trait first appearances tended to decrease with site age, with the exception of class 2. No last appearances occurred until class 3, where a single functional trait disappeared (Fig. 5a).

For spiders, the total number of functional traits followed a concave pattern, with the lower values in classes 0 and 4 and the higher values in classes 1, 2, and 3. Functional trait

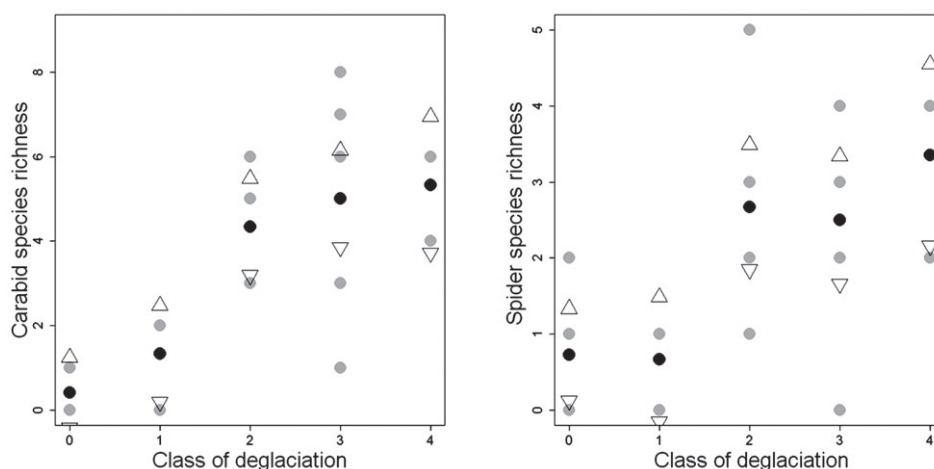


Fig. 2. Observed (grey dots) and expected (black dots) carabid and spider species richness in relation to the class of deglaciation. Triangles represent 95% confidence interval.

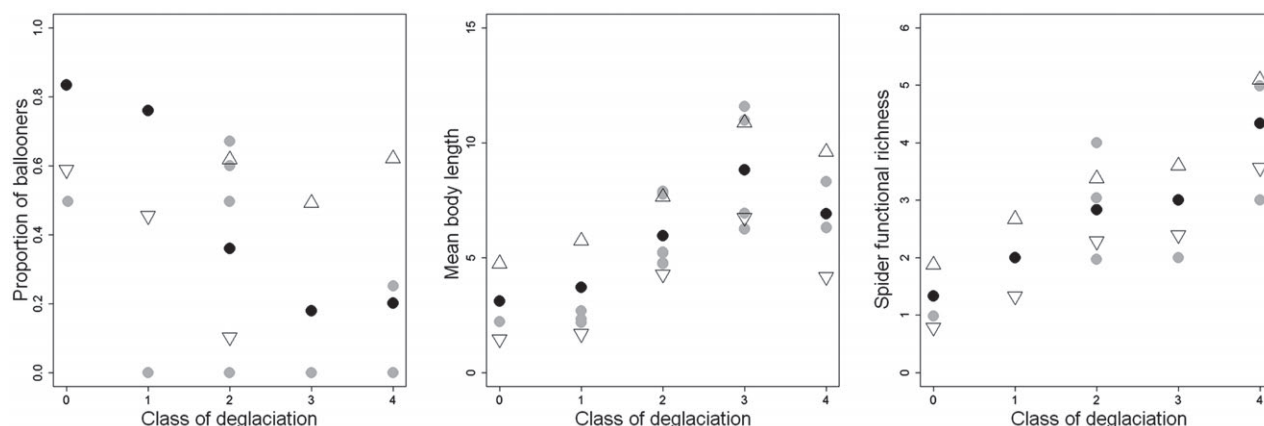


Fig. 3. Observed (grey dots) and expected (black dots) spider life-history traits in relation to the class of deglaciation. Only the cases in which the effect of time since deglaciation was supported (model including the variable more parsimonious than the null model; see text) are displayed. Triangles represent 95% confidence interval.

first appearances occur only within the first two classes of deglaciation, while the only last appearance occurred in class 3 (Fig. 5b).

Discussion

Species and their life-history traits on the debris-covered glacier

Debris-covered glaciers with their tongue descending below the treeline can host arthropod life on their surface (Gobbi *et al.*, 2011). Our study demonstrated that a debris-covered glacier with its tongue located above the treeline, is also capable of hosting arthropods. Specifically, we collected three different ground-dwelling arthropod species on the glacier: the carabid beetle *Nebria germari* Heer 1837, the wolf spider *Pardosa nigra* (C.L. Koch, 1834), and the linyphiid spider *Agyneta rurestris* (C. L. Koch, 1836). The life-history traits of these species are as follows: both *N. germari* and *P. nigra* are walking colonisers

with low dispersal ability, as the former is short-winged and the latter is not a ballooner, at least at the adult stage. Both species are ground hunters; specifically, *N. germari* is an olfactory-tactile predator (Brandmayr *et al.*, 2005), while *P. nigra* is a ground dweller with good eyesight which runs around in search of prey (Roberts, 1985). Notwithstanding the fact that these two species feed on similar prey, mainly collembolan and other insects (Raso *et al.*, 2014) transported as aeroplankton (Hagvar, 2012), the niche competition is reduced as they have different foraging habits: the former is a nocturnal predator (Homburg *et al.*, 2014) while the latter is mainly a diurnal predator (Raso *et al.*, 2014). Given the collection of juvenile instars on the glacier, it seems likely that both species complete their life cycle on the ice.

The spider *Agyneta rurestris* is widespread in Europe and its presence on the glacier is likely to be a result of its ability to quickly colonise pioneer habitats (Meijer, 1977). For this species, however, we have no evidence of its ability to reproduce on the glacier.

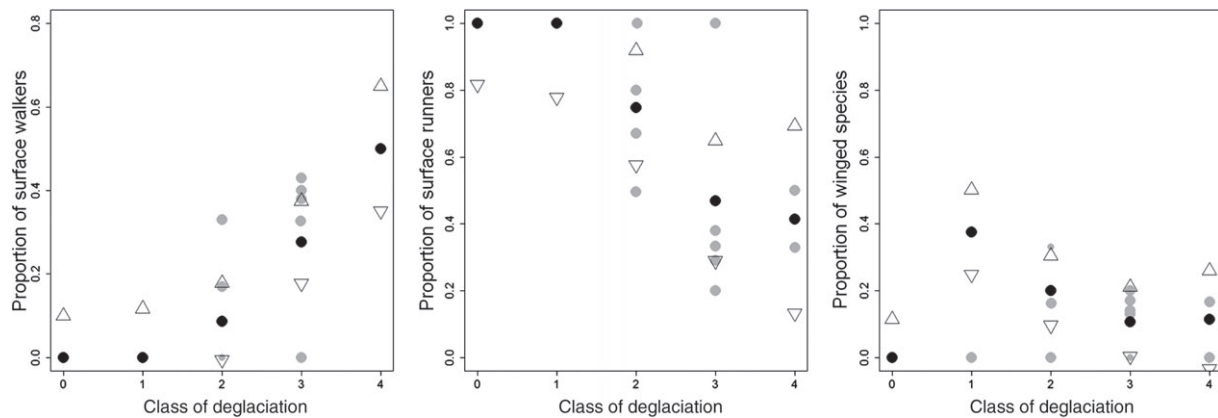


Fig. 4. Observed (grey dots) and expected (black dots) carabid life-history traits in relation to the class of deglaciation. Only the cases in which the effect of time since deglaciation was supported (model including the variable more parsimonious than the null model; see text) are displayed. Triangles represent 95% confidence interval.

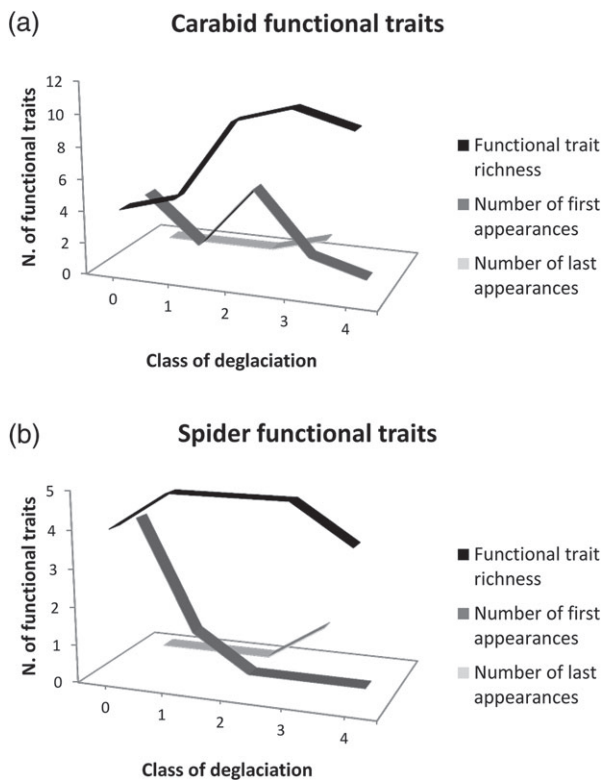


Fig. 5. Functional trait richness and functional trait first and last appearances among the classes of deglaciation: (a) carabids; (b) spiders.

Diversity

With respect to species richness values, differences among the five classes of deglaciation were found on both carabid beetles and spiders. The species richness pattern is in accordance with previous studies on invertebrate primary successions along glacier forelands (see Hagvar, 2012), confirming the increasing of number of species with the time since deglaciation.

In contrast, the functional diversity along the chronosequence of glacier retreat revealed different patterns in carabids when compared with spiders. Concerning carabids, no detectable trend was found, for either FRic or FDis. According to Mason *et al.* (2013) this result highlights that there is no change in influence of niche complementarity on either species occurrences or abundances, with increasing time since deglaciation. This result indicates that in our study system there are no habitat filtering processes (*sensu* HilleRisLambers *et al.*, 2012), and thus there are no environmental factors limiting the occurrence of species without certain traits. Interestingly, habitat filtering processes in carabid beetle distribution were found along glacier forelands located below the treeline due to a more complex habitat and community structure (Brambilla & Gobbi, 2014; Vater & Matthews, 2015). It is probably that, above the treeline the variation of complexity of habitat and community structure, in relation to the time since deglaciation, is not high enough to be able to filter the species/traits occurrence.

By contrast, for spiders, time since deglaciation positively affected the FRic, but not FDis. According to Mason *et al.* (2013), this result highlights an increasing influence of niche complementarity on species occurrences, but not abundances, with increasing time since deglaciation.

Life-history trait types and distribution

Trait distribution analysis revealed that on the glacier (class 0) and during the first stage of deglaciation (class 1), the early successional carabid assemblages were characterised by species with the following features: surface running larvae, mainly short-winged species, olfactory-tactil predators. Surface running larvae are probably mainly linked to the gravelly soils of the early successional stages, as they are effective at capturing their prey running between the stones or at the edge of the stones. According to this hypothesis, species with soil-exploring larvae (i.e. small larvae living in the soil; Brandmayr *et al.*, 2005) appeared in the mid- and late-successional stages, where the habitat maturity should sustain several prey species living

in the soil and with a low ability to escape (e.g. earthworms, fly larvae). After 20 years since glacier retreat (class 1–2) until the late successional stages (class 3–4), all larval hunting strategies (surface walkers, surface runners, soil pore explorers, spermatophagous), adult diet types (carnivorous, zoospermatophagous) and wing statuses (short-winged and winged) were represented and persisted along the glacier foreland. Therefore, this result supports the general pattern found in other glacier forelands where the number of low dispersal species increases in stable and mature environments (Gobbi *et al.*, 2007, 2010). Most of the sampled carabid species are olfactory-tactile hunters (Brandmayr *et al.*, 2005). This hunting strategy is considered to be the most primitive one, performed by unspecialised nocturnal predators with small eyes (Brandmayr *et al.*, 2005; Fountain-Jones *et al.*, 2015). As the olfactory-tactile hunting strategy is related to nocturnal predation (Brandmayr *et al.*, 2005), we can hypothesise that this strategy is particularly frequent in the species living in this kind of harsh habitat in order to partially avoid niche competition with spiders, and opiliones, which are also top predators (Hagvar, 2012), but with diurnal habits. Visual predators appeared only in late successional stages (classes 3–4). Visual hunting is typical of diurnal predators (e.g. *Notiophilus* spp.) with large eyes (Brandmayr *et al.*, 2005; Fountain-Jones *et al.*, 2015). Most carabids specialised in feeding on springtails occurring in late successional stages, where high vegetation cover favours high springtail abundance (Schirmel *et al.*, 2012). The omnivorous *Amara erraticus* (Duftschmid, 1812), the only seed-eating carabid, appears when vegetation cover is > 30%.

The analysis of spider trait distribution revealed that most of the hunting strategies are represented along the primary succession, but without a clear trend. The proportion of ballooners is higher on the glacier and in early successional stages, and then decreased along the succession. As a consequence, the dispersal strategy (ballooners versus ground dispersers) influenced spiders' distribution. Ballooning may be initiated by both environmental and physiological factors, and in general, overcrowding and food shortage can stimulate aerial dispersal (Duffey, 1998; Weyman *et al.*, 2002), which happens during the snow-free period (Coulson *et al.*, 2003). Spider body length increased along the primary succession. As bigger species are generally not ballooners, this trend can be explained by the correlation between body size and dispersal ability. In addition, our results are consistent with mechanisms invoking metabolic rate and desiccation resistance to predict an increase in body size from cool and moist habitats, such as the glacier surface and early successional stages, to warmer and dryer habitats, as late successional stages (Entling *et al.*, 2010).

In contrast to our expectations, we did not observe a true turnover of carabid and spider functional traits along the primary succession. Therefore, the presence of a filtering process on life-history traits can be excluded. Indeed, most of the traits were added and persisted, according to the 'addition and persistence model' (Vater, 2012; Vater & Matthews, 2013, 2015). Our results advance the hypothesis that, in our study system, the occurrence of ground beetles and spiders on the glacier and their distribution along the glacier foreland seem to be driven by dispersal ability and foraging strategies of each species.

Conclusions

Our results highlighted the fact that carabid and spider primary successions along a glacier foreland can be described not only by considering species diversity and turnover, as traditionally done, but also via the functional diversity and trait distribution approach, as already applied to plant assemblages (e.g. Caccianiga *et al.*, 2006; Erschbamer & Mayer, 2012). However, unlike plant assemblages, in our study system, carabid and spider species assemblages cannot be discriminated from their life-history trait types, as the traits are not mutually exclusive, but they mainly follow the 'addition and persistence model' and not the 'replacement change model' (Vater & Matthews, 2013). The 'addition and persistence model' of invertebrate succession consists of a process whereby, through time, taxa colonise deglaciated terrains without replacement of discrete communities (Vater, 2012). On the other hand, the proportion of most of the considered life-history traits within each species assemblage clearly changes in relation to the successional gradient; the species assemblages can thus be discriminated on the basis of the proportion of each trait. The use of life-history traits proved a useful tool to describe in more detail the ecological and behavioural features of the ground-dwelling arthropods involved in a primary succession triggered by glacier retreat.

To our knowledge, this is the first study to measure different components of functional diversity of ground-dwelling arthropods in response to glacier retreat and, in general, in harsh high-altitude environments. Using the trait-based approach and including functional diversity components, we contribute to the description of the adaptive strategies adopted by carabids and spiders colonising glacier surfaces and recently deglaciated terrains, landforms which are rapidly changing in response to the current global warming.

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final version of the manuscript. The authors declare that they do not have conflicts of interest.

Supporting Information

Additional Supporting Information may be found in the online version of this article under the DOI reference: 10.1111/een.12456

Figure S1. Thermal regime on the debris-covered glacier and on the glacier foreland over 1 year. The black line represents average values.

Table S1. Preliminary Spearman correlation analysis among the environmental covariates recorded in the study area.

References

- Azzoni, R.S., Franzetti, A., Fontaneto, D., Zullini, A. & Ambrosini, R. (2015) Nematodes and rotifers on two Alpine debris-covered glaciers. *Italian Journal of Zoology*, **82**, 616–623.
- Bartoń, K. (2016) *Package 'MuMIn'*. R package, version 1.15.6. R package [WWW document]. URL <https://CRAN.R-project.org/package=MuMIn> [accessed on 1 December 2016].
- Beale, C.M., Lennon, J.J., Yearsley, J.M., Brewer, M.J. & Elston, D.A. (2010) Regression analysis of spatial data. *Ecology Letters*, **13**, 246–264.
- Bell, J.R., Bohan, D.A., Shaw, E.M. & Weyman, G.S. (2005) Ballooning dispersal using silk: world fauna, phylogenies, genetics and models. *Bulletin of Entomological Research*, **95**, 69–114.
- Blandenier, G. (2009) Ballooning of spiders (Araneae) in Switzerland: general results from an eleven-year survey. *Arachnology*, **14**, 308–316.
- Brambilla, M. & Ficetola, G.F. (2012) Species distribution models as a tool to estimate reproductive parameters: a case study with a passerine bird species. *Journal of Animal Ecology*, **81**, 781–787.
- Brambilla, M. & Gobbi, M. (2014) A century of chasing the ice: delayed colonisation of ice-free sites by ground beetles along glacier forelands in the Alps. *Ecography*, **37**, 33–42.
- Brandmayr, P., Zetto Brandmayr, T. & Pizzolotto, R. (2005) *I Coleotteri Carabidi per la valutazione ambientale e la conservazione delle biodiversità*. Manuale operativo, Vol. 34. Agenzia per la Protezione dell'Ambiente e per i Servizi Tecnici, IGER, Roma, Italy.
- Burnham, K.P. & Anderson, D.R. (2002) *Model selection and multimodel inference: A practical information-theoretic approach*, 2nd edn. Springer-Verlag, New York, New York.
- Caccianiga, M., Luzzaro, A., Pierce, S., Cerabolini, B. & Ceriani, R.M. (2006) The functional basis of a primary succession resolved by CSR classification. *Oikos*, **112**, 10–20.
- Caccianiga, M., Andreis, C., Diolaiuti, G., D'Agata, C., Mihalcea, C. & Smiraglia, C. (2011) Alpine debris-covered glacier as a habitat for plant life. *The Holocene*, **21**, 1011–1020.
- Cardoso, P., Pekár, S., Jocqué, R. & Coddington, J.A. (2011) Global patterns of guild composition and functional diversity of spiders. *PLoS ONE*, **6**, e21710.
- Citterio, M., Diolaiuti, G., Smiraglia, C., D'Agata, C., Carnielli, T., Stella, G. *et al.* (2007) The fluctuations of Italian Glaciers during the last century: a contribution to knowledge about Alpine glacier changes. *Geografiska Annaler, Series A. Physical Geography*, **89**, 167–184.
- Coulson, S.J., Hodkinson, I.D. & Webb, N.R. (2003) Aerial dispersal of invertebrates over a high-Arctic glacier foreland: Midtre Lovenbreen, Svalbard. *Polar Biology*, **26**, 530–537.
- Díaz, S. & Cabido, M. (2001) Vive la différence: plant functional diversity matters to ecosystem processes. *Trends in Ecology and Evolution*, **16**, 646–655.
- Díaz, S., Purvis, A., Cornelissen, J.H.C., Mace, G.M., Donoghue, M.J. & Ewers, R.M. (2013) Functional traits, the phylogeny of function, and ecosystem service vulnerability. *Ecology and Evolution*, **3**, 2958–2975.
- Dormann, F., McPherson, C.M., Araujo, J.B., Bivand, M., Bolliger, R., Carl, G. *et al.* (2007) Methods to account for spatial autocorrelation in the analysis of species distributional data: a review. *Ecography*, **30**, 609–628.
- Duffey, E. (1998) Aerial dispersal in spiders. *Proceedings of the 17th European Colloquium of Arachnology, 1997* (ed. by P. A. Selden), pp. 187–191. British Arachnological Society, Edinburgh, U.K.
- Entling, W., Schmidt-Entling, M.H., Bacher, S., Brandl, R. & Nentwig, W. (2010) Body size–climate relationships of European spiders. *Journal of Biogeography*, **37**, 477–485.
- Erschbamer, B. & Mayer, R. (2012) Can successional species groups be discriminated based on their life history traits? A study from a glacier foreland in the Central Alps. *Plant Ecology & Diversity*, **4**, 341–351.
- Eymann, J., Degreef, J., Häuser, C., Monje, J.C., Samyn, Y. & VandenSpiegel, D. (2010) *Manual on Field Recording Techniques and Protocols for All Taxa Biodiversity Inventories and Monitoring*, Vol. 8, Part 2. Abc Taxa, U.K.
- Fountain-Jones, N.M., Baker, S.C. & Jordan, G.J. (2015) Moving beyond the guild concept: developing a practical functional trait framework for terrestrial beetles. *Ecological Entomology*, **40**, 1–13.
- Gobbi, M. & Brambilla, M. (2016) Patterns of spatial autocorrelation in the distribution and diversity of carabid beetles and spiders along Alpine glacier forelands. *Italian Journal of Zoology*, **83**, 600–605.
- Gobbi, M., De Bernardi, F., Pelfini, M., Rossaro, B. & Brandmayr, P. (2006) Epigeal arthropod succession along a 154 year glacier foreland chronosequence in the Forni Valley (Central Italian Alps). *Arctic, Antarctic and Alpine Research*, **38**, 357–362.
- Gobbi, M., Rossaro, B., Vater, A., De Bernardi, F., Pelfini, M. & Brandmayr, P. (2007) Environmental features influencing Carabid beetle (Coleoptera) assemblages along a recently deglaciated area in the Alpine region. *Ecological Entomology*, **32**, 682–689.
- Gobbi, M., Caccianiga, M., Cerabolini, B., Luzzaro, A. & De Bernardi, F. (2010) Plant adaptive response during primary succession are associated with functional adaptations in ground beetles on deglaciated terrain. *Community Ecology*, **11**, 223–231.
- Gobbi, M., Isaia, M. & De Bernardi, F. (2011) Arthropod colonization of a debris-covered glacier. *The Holocene*, **21**, 343–349.
- Hagvar, S. (2012) Primary succession in glacier forelands: how small animals conquer new land around melting glaciers. *International Perspective on Global Environmental Change* (ed. by S. S. Young and S. E. Silvern), pp. 151–172. INTECH, Rijeka, Croatia.
- HilleRisLambers, J., Adler, P.B., Harpole, W.S., Levine, J.M. & Mayfield, M.M. (2012) Rethinking community assembly through the lens of coexistence theory. *Annual Review of Ecology, Evolution, and Systematic*, **43**, 227–248.
- Homburg, K., Homburg, N., Schäfer, F., Schuldt, A. & Assmann, T. (2014) Carabids.Org—a dynamic online database of ground beetle species traits (Coleoptera, Carabidae). *Insect Conservation and Diversity*, **7**, 195–205.
- Hürka, K. (1996) *Carabidae of the Czech and Slovak Republics, České a Slovenské republiky*. Kabourek, Zlín, Czech Republic.
- Kaufmann, R. (2001) Invertebrate succession on an Alpine glacier foreland. *Ecology*, **82**, 2261–2278.
- Kotze, D.J., Brandmayr, P., Casale, A., Dauffy-Richard, E., Dekoninck, W., Koivula, M.J. *et al.* (2011) Forty years of carabid beetle research in Europe—from taxonomy, biology, ecology and population studies to

- bioindication, habitat assessment and conservation. *ZooKeys*, **100**, 55–148.
- Laliberté, E. & Legendre, P. (2010) A distance-based framework for measuring functional diversity from multiple traits. *Ecology*, **91**, 299–305.
- Laliberté, E., Legendre, P. & Shipley, B. (2014) *FD: Measuring Functional Diversity from Multiple Traits, and Other Tools for Functional Ecology*. R package, version 1.0-12.
- Legendre, P. & Legendre, L. (2012) *Numerical Ecology*. Elsevier, Amsterdam, The Netherlands.
- Losapio, G., Jordán, F., Caccianiga, M. & Gobbi, M. (2015) Structure-dynamic relationship of plant-insect networks along a primary succession gradient on a glacier foreland. *Ecological Modelling*, **314**, 73–79.
- Mason, N.W.H., Richardson, S.J., Peltzer, D.A., de Bello, F., Wardle, D.A. & Allen, R.B. (2012) Changes in coexistence mechanisms along a long-term soil chronosequence revealed by functional trait diversity. *Journal of Ecology*, **100**, 678–689.
- Mason, N.W.H., de Bello, F., Mouillot, D., Pavoine, S. & Dray, S. (2013) A guide for using functional diversity indices to reveal changes in assembly processes along ecological gradients. *Journal of Vegetation Science*, **24**, 794–806.
- Meijer, J. (1977) The immigration of spiders (Araneida) into a new polder. *Ecological Entomology*, **2**, 81–90.
- Moretti, M., Dias, A.T.C., de Bello, F., Altermatt, F., Chown, S.L., Azcárate, F.M. *et al.* (2017) Handbook of protocols for standardized measurement of terrestrial invertebrate functional traits. *Functional Ecology*, **31**, 558–567.
- Nentwig, W., Blick, T., Gloor, D., Hänggi, A. & Kropf, C. (2017) *Spiders of Europe*. Version 02.2017 [WWW document]. URL www.araneae.unibe.ch [accessed on 1 December 2016].
- Paul, F., Kääb, A. & Haeberli, W. (2007) Recent glacier changes in the Alps observed by satellite: consequences for future monitoring strategies. *Global and Planetary Change*, **56**, 111–122.
- Pesarini, C. & Monzini, V. (2010) Insetti della fauna italiana. Coleotteri Carabidi I. *Natura, Rivista di Scienze Naturali*, **100**, 152 pp.
- Pesarini, C. & Monzini, V. (2011) Insetti della fauna italiana. Coleotteri Carabidi II. *Natura, Rivista di Scienze Naturali*, **101**, 144 pp.
- Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D. & R Core Team. (2017) *nlme: Linear and Nonlinear Mixed Effects Models*. R package version 3.1-129 [WWW document]. URL <https://CRAN.R-project.org/package=nlme> [accessed on 1 December 2016].
- Pizzolotto, R., Albertini, A., Gobbi, M. & Brandmayr, P. (2016) Habitat diversity analysis along an altitudinal sequence of alpine habitats: the Carabid beetle assemblages as a study model. *Periodicum Biologorum*, **118**, 241–254.
- R Core Team (2016) *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. [WWW document]. URL <https://www.R-project.org/> [accessed on 1 December 2016].
- Raso, L., Sint, D., Mayer, R., Plangg, S., Recheis, R., Kaufmann, R. *et al.* (2014) Intraguild predation in pioneer predator communities of Alpine glacier forelands. *Molecular Ecology*, **23**, 3744–3754.
- Roberts, M.J. (1985) *The Spiders of Great Britain and Ireland*. Vol. 1. Atypidae to Theridiosomatidae. Harley Books, Colchester, U.K.
- Schirmel, J., Blindow, I. & Buchholz, S. (2012) Life-history trait and functional diversity patterns of ground beetles and spiders along a coastal heathland successional gradient. *Basic and Applied Ecology*, **13**, 606–614.
- Schleuter, D., Daufresne, M., Massol, F. & Argillier, C. (2010) A user's guide to functional diversity indices. *Ecological Monographs*, **80**, 469–484.
- Talarico, F., Giglio, A., Pizzolotto, R. & Brandmayr, P. (2016) A synthesis of feeding habits and reproduction rhythm in Italian seed-feeding ground beetles (Coleoptera: Carabidae). *European Journal of Entomology*, **113**, 325–336.
- Tampucci, D., Gobbi, M., Cabrini, E., Compostella, C., Marano, G., Pantini, P. *et al.* (2015) Plant and arthropod colonization of a glacier foreland in a peripheral mountain range. *Biodiversity*, **16**, 213–223.
- Tenan, S., Maffioletti, C., Caccianiga, M., Compostella, C., Seppi, R. & Gobbi, M. (2016) Hierarchical models for describing space-for-time variations in insect population size and sex-ratio along a primary succession. *Ecological Modelling*, **329**, 18–28.
- Vater, A.E. (2012) Insect and arachnid colonization on the Storbreen Glacier Foreland, Jotunheimen, Norway: persistence of taxa suggests an alternative model of succession. *The Holocene*, **22**, 1123–1133.
- Vater, A.E. & Matthews, J.A. (2013) Testing the 'addition 40 and persistence model' of invertebrate succession in a subalpine glacier-foreland Chronosequence: Fåbergstølsbreen, Southern Norway. *The Holocene*, **23**, 1151–1162.
- Vater, A.E. & Matthews, J.A. (2015) Succession of pitfall-trapped insects and arachnids on eight Norwegian glacier forelands along an altitudinal gradient: patterns and models. *The Holocene*, **25**, 108–129.
- Venables, W.N. & Ripley, B.D. (2002) *Modern Applied Statistics with S*. Springer, New York, New York.
- Vigna Taglianti, A. (2013) Fauna Europaea: Carabidae. *Fauna Europaea Version 2.6* (ed. by P. Audisio). [WWW document]. URL <http://www.faunaeu.org> [accessed on 1 April 2016].
- Walkley, A. & Black, I.A. (1934) An examination of the Degtjareff method for determining soil organic matter, and proposed modification of the chromic acid titration method. *Soil Science*, **37**, 29–38.
- Weyman, G.S., Sunderland, K.D. & Jepson, P.C. (2002) A review of the evolution and mechanisms of ballooning by spiders inhabiting arable farmland. *Ecology, Etology and Evolution*, **14**, 307–326.
- World Spider Catalog (2017) *World Spider Catalog. Version 18.0*. Natural History Museum, Bern, Switzerland. [WWW document]. URL <http://wsc.nmbe.ch> [accessed on 8 January 2017].

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